

Predicting Glucose–Insulin Dynamics

A Comparison of Physics-Informed Neural Networks
and Differentiable Physics

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Abstract

Glucose–insulin dynamics are central to understanding and managing diabetes, yet accurately modeling these processes remains a challenge. Hybrid approaches that combine mechanistic physiology with machine learning offer a promising pathway to improved glucose prediction. This study compares the predictive performance and modeling philosophy of two such approaches: a Physics-Informed Neural Network (PINN) and a Differentiable Physics (DP) model. Both are trained on 750 virtual patient trajectories generated by the FDA-accepted UVA/Padova Type 1 Diabetes Simulator, and both are built on the Bergman Minimal Model, but they differ in how physics is incorporated: the PINN imposes soft constraints through the loss function, whereas the DP model embeds the ODE solver directly in the learning loop. Within the PINN framework we evaluate RNN, LSTM, and Transformer architectures; the DP model employs a Neural ODE solver, with Flow Matching explored as an alternative. Predictive ability is quantified using the root mean square error (RMSE) between predicted and true blood glucose values. The LSTM-PINN attains the lowest error (0.495 mg/dL over a 15-minute horizon), while the Neural ODE recovers physiologically plausible Bergman parameters over full 72-hour trajectories, highlighting a trade-off between short-horizon accuracy and mechanistic faithfulness.

Keywords: glucose–insulin dynamics; physics-informed neural networks; differentiable physics; neural ODEs; Bergman Minimal Model; Type 1 diabetes.

1 Introduction

With the acceleration of applying machine learning, data, and AI in the world of science, specifically in biology, a fiercely debated question arises: does greater model complexity actually improve our ability to capture real physiological systems? Ultimately, our project is motivated by the desire to understand and compare two popular paradigms in biological system modeling: the flexibility and expressive power of “black-box” approaches versus the mechanistic structure and interpretability of “white-box” models.

To test our findings, we utilize the experimental ground of glucose–insulin dynamics, because they are the core physiological process behind diabetes control and present as a clean testbed for hybrid modeling that mixes known biology with learning from data. In this project, two collaborative teams experiment on different families of models, training and evaluating on the same dataset. The shared physical backbone across all models is a Bergman Minimal Model ODE system [1] that describes glucose, a remote insulin effect state, and insulin, with meals and exogenous insulin acting as inputs.

The PINN model [7] utilizes the Bergman Minimal Model ODE to contribute soft penalties to the neural network’s loss, penalizing the model for deviating from theoretical expectations. Differentiable physics takes the complementary path by embedding the full simulator inside the learning loop and making the solver differentiable, so that gradients can be back-propagated through the numerical integration

steps using automatic differentiation [2]. In the DP model, Bergman parameters are treated as trainable variables initialized from literature values and updated by back-propagating through a differentiable ODE integrator, while enforcing physiological plausibility through smooth constrained parameterizations and approximating the non-differentiable max term using ReLU.

Ultimately, this simultaneous approach allows us to shed light on the kind of complexity that is actually useful in biological modeling: whether it is flexibility that learns from the data, or theoretical structure that stays grounded in known physiology.

2 Methods

2.1 Dataset Generation

The dataset used in this study was generated using the UVA/Padova Type 1 Diabetes Simulator [3], an FDA-accepted physiological simulator widely used in diabetes research. The simulator produces physiologically realistic glucose and insulin trajectories for virtual patients under diverse meal consumption and insulin administration scenarios.

The dataset contains 750 simulated patient trajectories in total, of which 600 were used for training and 150 for validation. Each trajectory spans 72 hours sampled at 3-minute intervals, yielding 1,441 time steps per trajectory. The simulated population spans three distinct physiological cohorts — adults, adolescents, and children — each reflecting realistic variations in insulin sensitivity and glucose variability across age groups.

The two models consume this dataset differently. The PINN data loader extracts overlapping 60-step sliding windows from each trajectory, producing approximately 826,200 training samples of shape $(60, 3)$. The DP data loader instead extracts one complete 72-hour trajectory per patient–scenario pair in physical units, producing approximately 600 training samples of shape $(T,)$ where $T = 1441$ steps. Each DP sample contains the initial physiological state $s_0 = [G_0, X_0, I_0]$, the full meal disturbance sequence $D(t)$, the full insulin input sequence $u(t)$, and the observed blood glucose trajectory $G_{\text{true}}(t)$ used as supervision. The remote insulin compartment X_0 is initialized to zero, as it is latent and never directly observed. No normalization is applied to the DP data; the ODE integrator, Bergman parameters, and loss all operate in physical units (mg/dL, U/min) throughout.

2.2 Physics-Informed Neural Network

As a means of modeling glucose–insulin regulation among Type 1 diabetics, we developed a Physics-Informed Neural Network (PINN) framework, which integrates physiological knowledge with data-driven sequence modeling. Rather than relying solely on machine learning to determine metabolic dynamics, the PINN model incorporates specific mechanistic constraints derived from the Bergman Minimal Model of glucose–insulin regulation. The utility of this approach is to generate predictions that both accurately fit observed glucose trajectories and remain consistent with known physiological principles governing glucose metabolism.

2.2.1 Feature Selection

Three physiologic signals were incorporated as model inputs: blood glucose (BG), insulin delivery, and carbohydrate intake (CHO). Continuous glucose monitoring (CGM) measurements were not included, as CGM represents a noisy and time-lagged proxy for true glucose levels. Given that the simulator provides direct blood glucose measurements, inclusion of CGM data would potentially introduce additional noise without improving the physiological information available to the model. All other available columns are either derived from BG, which would introduce redundancy, or are risk indicators computed post hoc and are not independent physiological drivers.

2.2.2 Data Preprocessing

To maintain physiologically realistic values and remove potential simulator artifacts, blood glucose values were clipped to the range [30, 400] mg/dL. Missing insulin or carbohydrate values were imputed as zero. All input features were then independently standardized using statistics computed from the training dataset only:

$$\tilde{x} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}. \quad (1)$$

The same normalization parameters were applied to the validation set to prevent data leakage from the validation set into the normalization statistics.

2.2.3 Sliding Window Construction

To transform continuous trajectories into supervised learning samples, sliding temporal windows were extracted across each trajectory. Each model input consisted of a 60-step window corresponding to three hours of historical context, with shape (60, 3). The model predicted blood glucose over the next 5 time steps, corresponding to a 15-minute forecast horizon. The three-hour input window was chosen to capture the full metabolic response to a meal, including glucose absorption, insulin response, and post-prandial decay. The 15-minute prediction horizon matches forecasting intervals commonly used in artificial pancreas systems, where near-term glucose predictions guide insulin dosing decisions. To prevent information leakage, windows were extracted only within individual trajectories, ensuring samples never crossed patient or scenario boundaries.

2.2.4 Learning Framework

A PINN is a training paradigm in which the loss function contains two distinct components: a data loss measuring agreement between predictions and observed trajectories, and a physics loss penalizing violations of the governing equations. The total loss is defined as

$$L_{\text{total}} = L_{\text{data}} + \lambda L_{\text{physics}}, \quad (2)$$

where the hyperparameter λ controls the strength of the physics constraint relative to the data fit.

2.2.5 Governing Physiological Model

The physics constraints were derived from the Bergman Minimal Model, a classical three-compartment system describing glucose–insulin regulation:

$$\frac{dG}{dt} = -(p_1 + X)G + p_1 G_b + D(t), \quad (3)$$

$$\frac{dX}{dt} = -p_2 X + p_3 I, \quad (4)$$

$$\frac{dI}{dt} = -nI + \gamma \max(G - h, 0) + u(t), \quad (5)$$

where G is blood glucose concentration, X is the remote insulin effect, I is plasma insulin concentration, $D(t)$ is the carbohydrate absorption rate, and $u(t)$ is exogenous insulin administration. The physiological parameters were fixed to literature population averages rather than learned during training, distinguishing the PINN from the DP model, where these parameters are directly optimized.

2.2.6 Data Loss

The supervised component of the loss penalizes prediction errors in observable physiological variables:

$$L_{\text{data}} = \text{MSE}(G_{\text{pred}}, G_{\text{true}}) + \text{MSE}(I_{\text{pred}}, I_{\text{true}}). \quad (6)$$

Glucose and insulin predictions are weighted equally. The latent variable X is excluded from the data loss, as it is not directly observable.

2.2.7 Physics Loss

The physics loss enforces consistency between predicted trajectories and the Bergman differential equations. Since the neural network produces predictions at discrete time steps rather than continuous trajectories, time derivatives are approximated using forward finite differences:

$$\frac{dG}{dt} \approx \frac{G_{t+1} - G_t}{\Delta t}. \quad (7)$$

Residuals from each equation are computed by subtracting the model-implied derivative from the corresponding Bergman equation. As residuals approach zero, the predicted trajectories more fully satisfy the physiological model.

2.2.8 Physics Weight Calibration

To identify the optimal balance between data accuracy and physical consistency, a grid search was performed over

$$\lambda \in \{0.0, 0.001, 0.01, 0.1, 1.0\}. \quad (8)$$

A value of $\lambda = 0$ corresponds to a purely data-driven model with no physics regularization, while larger values enforce increasing adherence to the governing equations.

2.2.9 Latent Variable Modeling

The state variable X , representing the remote insulin effect, cannot be measured in clinical datasets and was treated as a latent variable. The network predicts X without direct supervision; its values are constrained only through the physics loss via Equation (4). This allows the model to learn an internal representation of insulin action that remains consistent with physiological dynamics.

2.2.10 Neural Network Architectures

Three neural network architectures were deployed within the same PINN framework to evaluate how different sequence modeling approaches interact with physics-derived learning: a recurrent neural network (RNN), a long short-term memory network (LSTM) [4], and a Transformer [8]. Their configurations are summarized in Table 1.

Table 1. Configuration of the three sequence architectures trained inside the PINN framework. All models take the same (60, 3) input window and are trained under the same loss in Equation (2).

Architecture	Layers	Width	Parameters
RNN	2 stacked recurrent layers	hidden size 128	$\approx 68,000$
LSTM	2 stacked LSTM layers	gated memory cells	$\approx 220,000$
Transformer	2 encoder layers	$d_{\text{model}} = 64$, 4 heads	$\approx 105,000$

The RNN-PINN serves as the baseline sequential architecture. It processes the 60-step input window sequentially, maintaining a continuously updating hidden state that summarizes past observations. Vanilla RNNs are known to suffer from vanishing gradients over long sequences, which is a meaningful limitation given our 60-step input window.

The LSTM-PINN extends the recurrent framework through gated memory cells that regulate information flow. These gates mitigate the vanishing gradient problem and allow the model to retain or forget information across longer time horizons. This architecture is well suited to glucose dynamics, where physiological response involves both slow basal metabolic processes and rapid post-prandial spikes.

The Transformer-PINN replaces recurrence with self-attention, enabling the model to evaluate relationships among all time steps simultaneously. Since the Transformer outputs a sequence of hidden

representations rather than a single state vector, mean pooling is applied across the sequence dimension to produce a fixed-length representation before generating predictions.

All three architectures were trained under the same PINN loss function, so differences in performance can be attributed to architectural properties rather than changes in the physics-informed objective.

2.2.11 Evaluation Metric

Model performance was evaluated using the root mean square error (RMSE) between predicted and true blood glucose values:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum (G_{\text{pred}} - G_{\text{true}})^2}. \quad (9)$$

RMSE was computed after converting predictions back to physiological units (mg/dL) to ensure clinical interpretability. Lower RMSE indicates predicted glucose trajectories more closely align with the underlying physiological dynamics.

2.3 Differentiable Physics

The Differentiable Physics (DP) model differs significantly from the PINN models. Even though both utilize the same Bergman Minimal Model equations, the DP model is structurally different for three reasons.

1. **Physics is structural, not a penalty.** The ODE solver is placed inside the computation graph. The model cannot output anything that is not a solution to the (augmented) Bergman equations, so violations are impossible by construction.
2. **The task is parameter calibration over full trajectories, not short-horizon forecasting.** The DP model receives an initial physiological state and integrates the ODE forward over a full 72-hour trajectory. Physiological rate constants cannot be reliably identified from a 15-minute window; the full system response is required.
3. **Bergman parameters are learned from data rather than fixed.** The five parameters $[p_1, p_2, p_3, n, \gamma]$ are trainable. After training, their values can be compared directly to known physiological ranges to assess model calibration.

The Neural ODE was the primary DP model, built on the Bergman Minimal Model augmented with a learned correction term. With state vector $s = [G, X, I]^\top$,

$$\frac{dG}{dt} = -(p_1 + X)G + p_1 G_b + D(t) + f_\theta(G, X, I, t), \quad (10)$$

$$\frac{dX}{dt} = -p_2 X + p_3 I, \quad (11)$$

$$\frac{dI}{dt} = -nI + \gamma \max(G - h, 0) + u(t). \quad (12)$$

The blue term $f_\theta(G, X, I, t)$ is the learned neural correction. The dX/dt and dI/dt equations are kept in pure Bergman form because insulin and remote insulin dynamics are well captured by the exponential decay structure; only the glucose equation is augmented.

2.3.1 Learned Components

The Neural ODE has two sets of trainable parameters. The first is a set of global Bergman parameters $\theta = [p_1, p_2, p_3, n, \gamma]$, five scalar values shared across all patients. These are stored as unconstrained values $\tilde{\theta}$ and mapped to positive physiological values via softplus:

$$\theta_i = \text{softplus}(\tilde{\theta}_i) = \log(1 + e^{\tilde{\theta}_i}) > 0. \quad (13)$$

After softplus, a hard clamp enforces physiological bounds $p_1, p_2 \in [10^{-4}, 10^{-1}]$, $p_3 \in [10^{-8}, 10^{-3}]$, $n \in [10^{-4}, 0.5]$, and $\gamma \in [10^{-4}, 0.05]$:

$$\theta_i \leftarrow \text{clamp}(\theta_i, \theta_i^{\min}, \theta_i^{\max}). \quad (14)$$

The unconstrained values $\tilde{\theta}$ are initialized such that $\text{softplus}(\tilde{\theta}_i) = \theta_i^{\text{lit}}$ at epoch 0, meaning training begins exactly at the Bergman literature parameter values.

The second learned component is the MLP augmentation $f_\theta : \mathbb{R}^4 \rightarrow \mathbb{R}$, a deliberately small two-layer network with 32 hidden units and tanh activations. The final layer is initialized to zero, so at epoch 0 the model reduces exactly to the pure Bergman equations. The MLP is kept small so that the ODE structure carries most of the explanatory burden.

2.3.2 Forward Pass

Given a batch of initial states $s_0 \in \mathbb{R}^{B \times 3}$ and input sequences $D, u \in \mathbb{R}^{B \times T}$, the augmented Bergman equations are integrated forward using a fixed-step RK4 solver:

$$s(t) = s_0 + \int_0^T f_{\text{aug}}(s(\tau), \theta, D(\tau), u(\tau)) d\tau. \quad (15)$$

The predicted glucose trajectory is extracted as $\hat{G}(t) = s(t)_G$, giving shape (B, T) .

2.3.3 Training

The training loss is the mean squared error between the predicted and observed glucose trajectory over the full trajectory:

$$\mathcal{L}_{\text{NODE}} = \frac{1}{BT} \sum_{b=1}^B \sum_{t=1}^T \left(\hat{G}_b(t) - G_b^{\text{true}}(t) \right)^2. \quad (16)$$

No physics penalty is required, since any output is by construction a solution to the augmented Bergman equations. Both θ and the MLP weights are updated jointly using Adam [5] with learning rate 10^{-3} and cosine annealing to 10^{-5} . Gradients are clipped to a maximum ℓ_2 norm of 1.0 to prevent instability during early training.

3 Results

3.1 Physics-Informed Neural Network

Three architectures (RNN, LSTM, Transformer) were each trained for 50 epochs across five physics-loss weights $\lambda \in \{0.0, 0.001, 0.01, 0.1, 1.0\}$, yielding 15 distinct models in total. Table 2 reports the best validation RMSE for each combination.

Table 2. Best validation RMSE (mg/dL) by architecture and physics-loss weight λ . Bold indicates the best result per architecture.

	$\lambda = 0$	$\lambda = 0.001$	$\lambda = 0.01$	$\lambda = 0.1$	$\lambda = 1.0$
RNN	0.693	0.712	0.726	0.797	1.520
LSTM	0.498	0.516	0.495	0.571	1.412
Transformer	3.112	2.677	2.296	2.563	3.992

The LSTM achieved the lowest overall RMSE at 0.495 mg/dL with $\lambda = 0.01$. The effect of the physics-loss term varied across architectures. For the LSTM, the RMSE difference between $\lambda = 0$ and

$\lambda = 0.01$ was only 0.003 mg/dL, which is practically negligible. For the Transformer, the reduction from $\lambda = 0$ to $\lambda = 0.01$ was more substantial, suggesting that physics regularization provided more meaningful guidance for the weaker architecture. The RNN showed very little benefit from physics regularization. As shown in Figure 1, all three architectures degraded sharply at $\lambda = 1.0$, with RMSE roughly doubling across the board.

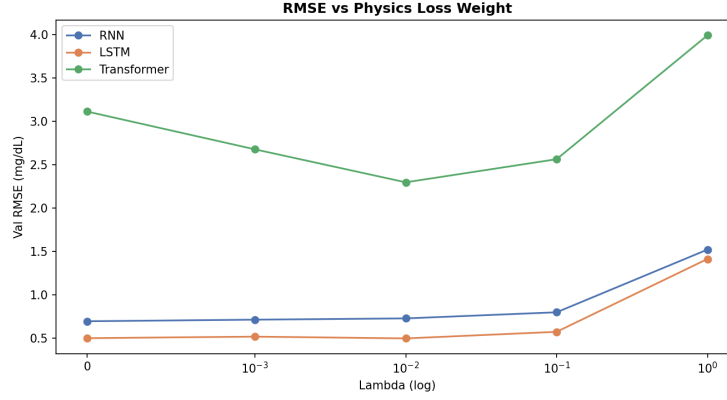


Figure 1. Validation RMSE versus physics-loss weight λ for all three PINN architectures.

The LSTM-PINN and RNN-PINN training curves follow a similar trajectory (Figure 2), corroborating their relatively similar RMSE scores, with data loss consistently lying below physics loss in both models. The Transformer model stands out: the RMSE curves are more volatile across all λ values, and the best RMSE was achieved much earlier in training, suggesting that with too many epochs the Transformer begins to overfit.

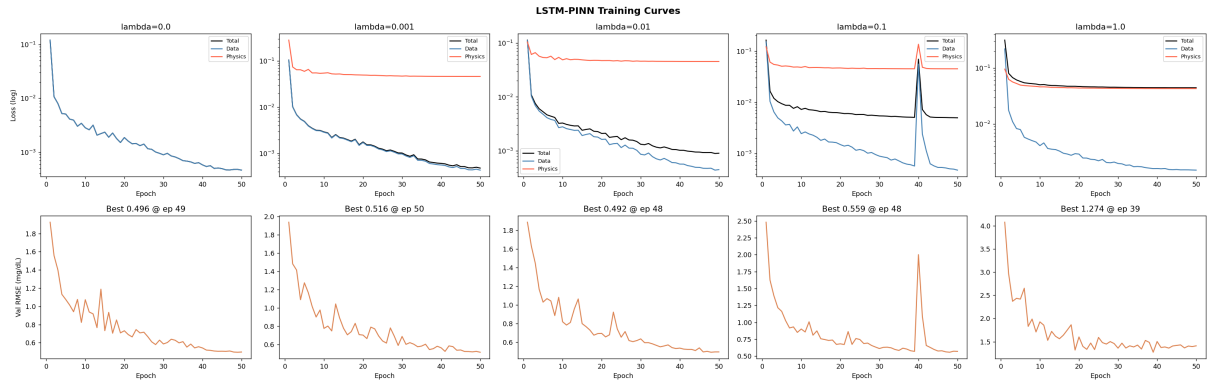
3.2 Differentiable Physics

The Neural ODE was trained for 50 epochs, achieving a best validation RMSE of 43.997 mg/dL. As shown in ??, the MSE loss dropped sharply in the first few epochs before stabilizing, and the validation RMSE followed a similar pattern, converging quickly and remaining stable through the remainder of training. The parameter convergence plot shows that both p_1 and n moved away from their literature initializations over the course of training, with n converging above its literature value and p_1 converging slightly above its own.

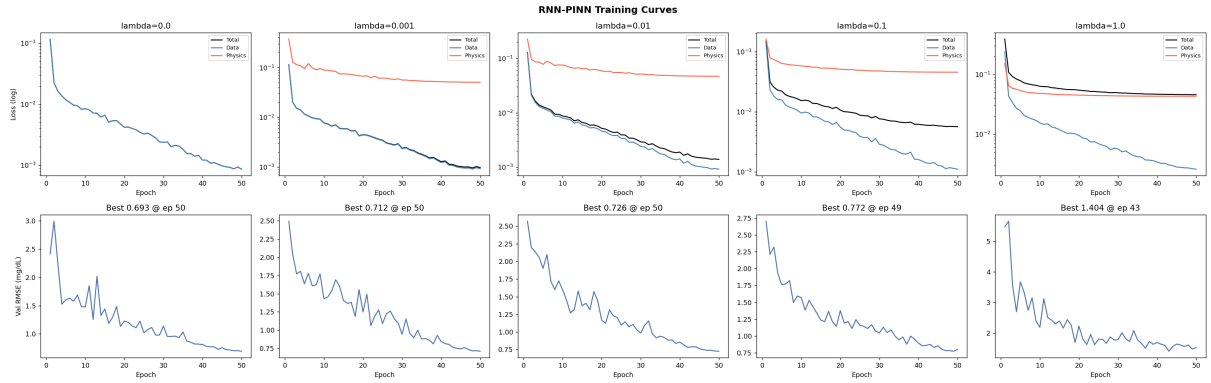
Table 3 reports the full set of learned Bergman parameters after training, compared to their literature values. All five parameters remained within a factor of 1.25 $\times$ of their literature values, suggesting that the model successfully calibrated physiologically plausible parameters from data rather than diverging to implausible values.

Table 3. Learned versus literature Bergman parameters after 50 epochs of Neural ODE training.

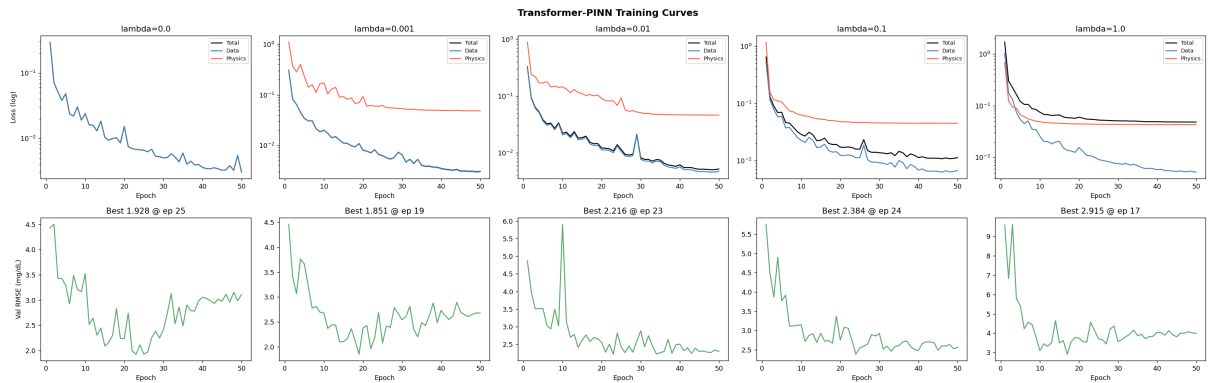
Parameter	Learned	Literature	Ratio
p_1 (glucose effectiveness)	0.034535	0.028000	1.23 $\times$
p_2 (remote insulin decay)	0.029604	0.025000	1.18 $\times$
p_3 (insulin action on glucose)	0.000048	0.000050	0.96 $\times$
n (insulin clearance rate)	0.104884	0.090000	1.17 $\times$
γ (pancreatic secretion gain)	0.006461	0.006000	1.08 $\times$



(a) LSTM-PINN



(b) RNN-PINN



(c) Transformer-PINN

Figure 2. Training curves for all three PINN architectures across $\lambda \in \{0.0, 0.001, 0.01, 0.1, 1.0\}$. Top row of each panel: loss components (log scale). Bottom row: validation RMSE.

4 Discussion

4.1 Physics-Informed Neural Network

The PINN results suggest that incorporating physics into the learning process was helpful, but its benefit was modest and architecture-dependent. Among the architectures tested, the LSTM performed best overall, while the RNN was less effective and the Transformer performed worst in this setting. Because all three models were trained under the same physics-informed framework, this difference likely reflects how well each architecture captures the temporal structure of glucose–insulin regulation rather than differences in the mechanistic formulation itself. In particular, the LSTM’s stronger sequential memory appears better suited to this task, where current glucose levels depend not only on immediate inputs but also on delayed physiological effects over time.

Another important finding is that the best performance occurred at a small nonzero physics-loss weight rather than at zero. This indicates that the physics term provided useful regularization, since a modest amount of mechanistic guidance slightly improved performance compared with a purely data-driven formulation. At the same time, performance worsened as the physics term became too strong, revealing the central trade-off in the PINN approach: the physics term can improve generalization when it acts as a gentle constraint, but if it dominates the objective, the model becomes too rigid and loses the flexibility needed to fit complex trajectories.

It is important to note that the PINN’s low RMSE values should be interpreted carefully. For a 15-minute forecast with three hours of input context, the model may have primarily learned short-term temporal persistence rather than genuine physiological dynamics. The low RMSE here does not necessarily mean the model has learned physiology; rather, it has likely learned to predict small, smooth fluctuations in glucose that are predictable from recent history alone.

The RNN’s monotonic degradation with increasing λ is also notable. The physics regularization only helped architectures that were already struggling (Transformer), or barely helped the model that was already performing well (LSTM). For a competent sequence model at this prediction horizon, the physics constraint appears to have had little practical relevance.

Finally, the LSTM’s advantage over the Transformer can be understood through the nature of the task. With a short 15-minute prediction horizon and smooth inter-meal dynamics, the LSTM only needs to carry information forward for 60 steps. Transformer models are designed for global attention, which is likely overkill here: for glucose dynamics, the most recent 10–20 minutes dominate the near-term forecast, and the Transformer’s tendency to attend to distant timestamps equally may actually hurt more than help.

4.2 Differentiable Physics

The differentiable physics approach provides a more mechanistically grounded way to model glucose dynamics, as the Bergman system is incorporated directly into the learning process rather than as a soft penalty. The Neural ODE uses integration of the dynamical system to generate trajectories, making the model structurally consistent with physiological assumptions by construction.

One of the major advantages of this approach is interpretability. Because the Neural ODE includes both the latent physiological state and the Bergman-related parameters, the model functions not only as a predictor but as a calibration tool for glucose and insulin dynamics. The learned parameters carry direct physiological meaning and can be examined to determine whether they fall within plausible ranges, which is not possible for a black-box forecasting model.

The trajectory-level formulation of the DP model is also noteworthy. Unlike the PINN, where prediction is limited to a 15-minute horizon with a fresh window of ground truth provided at each step, the Neural ODE integrates forward over 72 hours from a single initial state. This is a substantially harder task. Small discrepancies in the initial stages of integration compound over time, and the model has no opportunity

to correct against observed data mid-trajectory. The DP model should therefore be understood as a physiological simulator rather than a short-term forecaster, and RMSE alone is insufficient to compare it to the PINN on equal terms.

The results also highlight an important limitation of the current implementation. The mechanistic structure improves physiological coherence but may restrict flexibility when glucose trajectories exhibit sharp meal-driven excursions or rapid recovery phases. The Bergman ODE was originally developed to capture population-level glucose–insulin dynamics, and the simulated dataset may reflect more specific patterns that the simple augmented formulation cannot fully capture. Future improvements could include richer physiological structure, better handling of sharp exogenous disturbances, or hybrid approaches that preserve mechanistic understanding while allowing greater representational flexibility.

Although Flow Matching [6] was explored as an additional trajectory-generation experiment, it is not treated here as the primary DP result. Unlike the Neural ODE, it does not use the Bergman equations as governing dynamics, and is better understood as an exploratory extension rather than a differentiable physics model in the strict sense.

4.3 Comparing the Two Paradigms

The comparison between the two model families reveals an important trade-off between prediction accuracy and physiological faithfulness, summarized in Table 4. In terms of raw performance, the PINN achieved substantially lower RMSE than the DP model, with the LSTM-PINN performing best overall. However, this comparison is not straightforward. The PINN is reset with three hours of ground truth at every prediction window, making it difficult for errors to compound. The DP model, by contrast, integrates freely from a single initial state over 72 hours with no such correction. These are fundamentally different tasks, and RMSE alone does not capture this asymmetry.

Table 4. Summary of the two modeling paradigms. The reported RMSE values correspond to different prediction tasks and are therefore not directly comparable.

	PINN	Differentiable Physics
Role of physics	Soft penalty in the loss	Structural, inside the solver
Bergman parameters	Fixed to literature values	Learned from data
Prediction task	15-minute forecast	72-hour free integration
Ground truth at test	Refreshed every window	Initial state only
Best validation RMSE	0.495 mg/dL (LSTM)	43.997 mg/dL (Neural ODE)
Interpretability	Largely black-box	Explicit physiological states

The DP model is more interpretable and more physiologically governed. Its predictions are tied to explicit physiological states and learned parameters that carry direct mechanistic meaning. The PINN is physics-informed in the sense that it adds a mechanistic penalty to an otherwise flexible neural network, but the network itself remains largely black-box. The DP model is also more flexible after training in one important sense: because it produces parameters for explicit equations, the governing equations can be evaluated continuously and used to generate predictions at arbitrary time points. The PINN’s predictions, by contrast, are constrained to the specific windowed inputs it was trained on.

5 Conclusion

The central trade-off of this project can be stated directly: the PINN gives better forecasting accuracy but uses physics in a weaker way, while the DP model offers stronger mechanistic grounding and interpretability at the cost of higher prediction error at long horizons. The most promising direction going forward may be

models that combine the predictive flexibility of the PINN with the structural guarantees of differentiable physics.

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